1_Topic_LogicReview2

Zadanie	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Maks. pkt	1.5	1.5	1.5	1.5	1.5	1.5	2.5	1.5	1.5	6	1.5	3.5	1.5	3.5	
Zdobyte															
Zadanie	**15**	**16**													**Suma**
Maks. pkt	3	3.5													37
Zdobyte															

Zadanie 1: (1.5 pkt)

1.1 Calculate, according to truth - tables:

c) $[(\mathbf{1} \Rightarrow \mathbf{0}) \Rightarrow \mathbf{1}] \Rightarrow (\mathbf{1} \Rightarrow \mathbf{0})$,

Zadanie 2: (1.5 pkt)

1.6 The same claim with variables varied through set $\mathbb{R}$:

b) *the product of two numbers with different signs is a positive number* $(<, \cdot, 0)$,

Zadanie 3: (1.5 pkt)

1.3 For how many assignments the formula is true?

o) $(p \Rightarrow q) \Rightarrow [(q \Rightarrow r) \Rightarrow (p \Rightarrow r)]$.

Zadanie 4: (1.5 pkt)

1.8 Interpret symbols $\alpha, *$ occurring in following formulas to make them (1) true, (2) false.

d) $(\exists x)(\forall y)x * y = y$

Zadanie 5: (1.5 pkt)

1.2 Let $x, y, z, t \in \{\mathbf{0}, \mathbf{1}\}$. Solve the following equations:

c) $\sim [(x \vee y \vee t) \Rightarrow (\sim x \vee z \vee \sim t)] = \mathbf{0}$

Zadanie 6: (1.5 pkt)

1.1 Calculate, according to truth - tables:

c) $[(\mathbf{1} \Rightarrow \mathbf{0}) \Rightarrow \mathbf{1}] \Rightarrow (\mathbf{1} \Rightarrow \mathbf{0})$,

Zadanie 7: (2.5 pkt)

1.5 Write the mathematical formulas corresponding to the following statements with the help of the following signs only: propositional connectives, quantifiers, variables varied through set $\mathbb{N}$ and symbols indicated in brackets (define auxiliary symbols if necessary)

a) *x is a divisor of y* (symbols: $=, <, \cdot$),

Zadanie 7 (cd.):

l) *there exists one and only one prime number between 10 and 20* $(=, <, \cdot, 1)$,

Zadanie 8: (1.5 pkt)

1.3 For how many assignments the formula is true?

l) $(p \wedge q \wedge \sim r) \Leftrightarrow [(\sim p \Rightarrow \sim q) \vee r]$

Zadanie 9: (1.5 pkt)

1.8 Interpret symbols $\alpha, *$ occurring in following formulas to make them (1) true, (2) false.

d) $(\exists x)(\forall y) x * y = y$

Zadanie 10: (6 pkt)

1.5 Write the mathematical formulas corresponding to the following statements with the help of the following signs only: propositional connectives, quantifiers, variables varied through set $\mathbb{N}$ and symbols indicated in brackets (define auxiliary symbols if necessary)

a) x *is a divisor of* y (symbols: $=, <, \cdot$),

b) x *is even* $(=, +)$,

Zadanie 10 (cd.):

i) *there doesn't exist the greatest number* $(\leq)$,

m) *there are no numbers with at most two multiples being squares* $(=, <, \cdot)$,

p) *the product of prime numbers is a sum of three prime numbers* $(=, <, \cdot, +, 1)$,

Zadanie 11: (1.5 pkt)

Define the connectives: negation, disjunction, conjunction and implication in terms of

b) Peirce's functor $\downarrow$, where $p \downarrow q \Leftrightarrow \sim p \wedge \sim q$.

Zadanie 12: (3.5 pkt)

1.8 Interpret symbols $\alpha, *$ occurring in following formulas to make them (1) true, (2) false.

b) $(\forall x, y, z)[\alpha(x, y) \wedge \alpha(y, z) \Rightarrow \alpha(x, z)]$

c) $(\forall x, y)\alpha(x, y) \Leftrightarrow (\forall x)\alpha(x, x)$

d) $(\exists x)(\forall y)x * y = y$

Zadanie 13: (1.5 pkt)

1.2 Let $x, y, z, t \in \{\mathbf{0}, \mathbf{1}\}$. Solve the following equations:

d) $(x \Leftrightarrow y) \wedge (y \Leftrightarrow\sim z) \wedge (z \Leftrightarrow\sim t) = \mathbf{1}$,

Zadanie 14: (3.5 pkt)

1.3 For how many assignments the formula is true?

a) $[p \vee (q \Rightarrow r)] \wedge [\sim r \vee (q \Rightarrow p)]$

n) $[(p \Rightarrow q) \Rightarrow r] \vee [(\sim r \vee q) \wedge r]$,

o) $(p \Rightarrow q) \Rightarrow [(q \Rightarrow r) \Rightarrow (p \Rightarrow r)]$.

Zadanie 15: (3 pkt)

1.6 The same claim with variables varied through set $\mathbb{R}$:

e) *each linear equation has a unique solution* $(=, 0, \cdot, +)$,

Zadanie 15 (cd.):

f) *there exist a quadratic polynomial with exactly two solutions* $(=, 0, \cdot, +)$,

g) *each system of two linear equations with two variables has a unique solution* $(=, 0, \cdot, +)$,

i) *not every real is a value of a quadratic polynomial with positive coefficients* $(=, +, \cdot, 0, >)$.

Zadanie 16: (3.5 pkt)

1.1 Calculate, according to truth - tables:

b) $(\mathbf{0} \Rightarrow \mathbf{0}) \Rightarrow [\mathbf{0} \Rightarrow (\mathbf{0} \Rightarrow \mathbf{0})]$,

c) $[(\mathbf{1} \Rightarrow \mathbf{0}) \Rightarrow \mathbf{1}] \Rightarrow (\mathbf{1} \Rightarrow \mathbf{0})$,

e) $(\mathbf{1} \Rightarrow \mathbf{0}) \Leftrightarrow [(\mathbf{0} \Rightarrow \mathbf{1}) \vee \mathbf{0}]$.